HDFS & Distributed Systems – MCQ Quiz

1. What does HDFS stand for?
   a) High Distribution File System
   b) Hadoop Distributed File System
   c) Hierarchical Data File System
   d) Hadoop Data Framework System

   Ans a)

2. In HDFS, what is the default block size in Hadoop 2.x and later?
   a) 64 MB
   b) 32 MB
   c) 128 MB
   d) 256 MB

   Ans c)

3. Which component of HDFS is responsible for storing metadata about files and blocks?
   a) DataNode
   b) NameNode
   c) Secondary NameNode (StandBy namenode) HA mode (High Availability)
   d) ResourceManager

   Ans b)

4. What is the default replication factor in HDFS?
   a) 1
   b) 2
   c) 3
   d) 5

   Ans c)

5. What is the primary role of the Secondary NameNode?
   a) Acts as a backup NameNode for failover
   b) Periodically merges the FsImage and EditLogs
   c) Stores actual data blocks
   d) Manages resource allocation

   Ans b)

6. Which of the following best describes the CAP theorem?
   a) A system can guarantee Consistency, Availability, and Partition tolerance simultaneously
   b) A distributed system can only guarantee at most two of Consistency, Availability, and Partition tolerance
   c) A system must sacrifice Consistency for scalability
   d) CAP applies only to relational databases

   Ans a)

7. In HDFS architecture, what is a "rack awareness" policy used for?
   a) Encrypting data across racks
   b) Optimizing network bandwidth and improving fault tolerance by considering node

placement
c) Balancing CPU load across nodes
d) Managing user authentication

Ans b)

8. Which protocol does HDFS primarily use for communication between client and NameNode?
a) FTP
b) HTTP only
c) RPC (Remote Procedure Call) (SSH)
d) SMTP

Ans c)

9. What happens when a DataNode fails in HDFS?
a) The entire cluster stops
b) The NameNode detects it via missed heartbeats and re-replicates blocks to other DataNodes
c) Data is permanently lost
d) The Secondary NameNode takes over as DataNode

Ans b)

10. In distributed systems, what is "eventual consistency"?
a) All nodes are always consistent immediately
b) Given enough time without new updates, all replicas will converge to the same value
c) Consistency is never achieved
d) Only the primary node needs to be consistent

Ans b)

11. What is the purpose of the "heartbeat" mechanism in HDFS?
a) To encrypt data transfers
b) To signal that a DataNode is alive and functioning to the NameNode
c) To balance load between clients
d) To trigger block replication manually

Ans b)

12. Which of the following is a single point of failure in classic (non-HA) HDFS architecture?
a) DataNode
b) NameNode
c) Secondary NameNode
d) Client

Ans b)

13. What is "write-once-read-many" (WORM) in the context of HDFS?
a) A security protocol
b) A file access model where files are written once and read multiple times without modification
c) A backup strategy
d) A replication algorithm

Ans b)

*In distributed systems, what does "sharding" refer to?
a) Encrypting data across nodes
b) Partitioning data across multiple servers/nodes to distribute load
c) Replicating entire datasets on every node
d) Compressing data before storage

14. What is the main purpose of the "Paxos" or "Raft" algorithms in distributed systems?
    a) Data compression
    b) Achieving consensus among distributed nodes
    c) Load balancing
    d) Data encryption

    Ans b)

15. Which HDFS feature allows high availability by running two NameNodes (active and standby)?
    a) Rack Awareness
    b) HDFS Federation
    c) HDFS High Availability (HA) with QJM (Quorum Journal Manager)
    d) Secondary NameNode

    Ans c)

16. What is a "split-brain" scenario in distributed systems?
    a) When data is split across multiple racks
    b) When two nodes both believe they are the leader/master simultaneously, causing inconsistency
    c) When a file is divided into blocks
    d) When replication fails

    Ans b)

17. In HDFS, which command is used to check the health of the filesystem?
    a) hdfs dfs -health
    b) hdfs fsck
    c) hdfs dfs -status
    d) hdfs check -all

    Ans b)

18. What is the primary trade-off addressed by the "quorum-based" replication approach in distributed systems?
    a) Balancing consistency and availability by requiring a majority of nodes to agree
    b) Reducing storage costs
    c) Increasing encryption strength
    d) Simplifying client APIs

    Ans a)

19. Which of the following is NOT a characteristic of HDFS?
    a) Designed for large files

b) Optimized for low-latency random access to small files
c) Fault-tolerant through replication
d) Follows a master-slave architecture

Ans b)

Answers

1. b) Hadoop Distributed File System
2. c) 128 MB
3. b) NameNode

4. c) 3
5. b) Periodically merges the FsImage and EditLogs
6. b) A distributed system can only guarantee at most two of Consistency, Availability, and Partition tolerance
7. b) Optimizing network bandwidth and improving fault tolerance by considering node placement
8. c) RPC (Remote Procedure Call)
9. b) The NameNode detects it via missed heartbeats and re-replicates blocks to other DataNodes
10. b) Given enough time without new updates, all replicas will converge to the same value
11. b) To signal that a DataNode is alive and functioning to the NameNode
12. b) NameNode
13. b) A file access model where files are written once and read multiple times without modification
14. b) Partitioning data across multiple servers/nodes to distribute load
15. b) Achieving consensus among distributed nodes
16. c) HDFS High Availability (HA) with QJM (Quorum Journal Manager)
17. b) When two nodes both believe they are the leader/master simultaneously, causing inconsistency
18. b) hdfs fsck
19. a) Balancing consistency and availability by requiring a majority of nodes to agree
20. b) Optimized for low-latency random access to small files